

Gold Future/ETF Analysis

Joe Hollander

2025

Table 1: A summary of the regression coefficients and measures of goodness of fit for regressing one-day returns of 1, 2, 6 and 12-month futures on 1-day returns of spot gold from 12/22/2008 to 7/14/2014.

Response	Slope	Intercept	R^2	RMSE
1-Month	0.94314	2.419×10^{-5}	0.80947	0.00530
2-Month	0.94301	$\approx 10^{-5}$	0.80911	0.00530
6-Month	0.94348	3.180×10^{-5}	0.80934	0.00530
12-Month	0.94358	-5.233×10^{-5}	0.80984	0.00529

Table 2: A summary of the slopes and R^2 from the regressions of futures returns versus gold returns over different holding periods.

	Days	1-Month	2-Month	6-Month	12-Month
Slope	1	0.94314	0.94301	0.94348	0.94358
	5	0.99805	0.99752	0.99757	0.99715
	10	1.01075	1.01020	1.00970	1.00841
	15	0.99448	0.99461	0.99431	0.99345
R^2	1	0.80947	0.80911	0.80934	0.80984
	5	0.97154	0.97158	0.97194	0.97165
	10	0.98516	0.98530	0.98572	0.98550
	15	0.98704	0.98691	0.98725	0.98713

Table 3: A summary of the weights and in/out of sample RMSEs for portfolios of 1, 2, and 3 futures contracts.

Portfolio		w_0	w_1	w_2	w_3	RMSE (in)	RMSE (out)
1 Futures	1-m	-0.01071	1.01071			6.62989	3.34047
	2-m	-0.04835	1.04835			12.22148	6.12074
	6-m	-0.05030	1.05030			13.23663	5.33971
	12-m	-0.06842	1.06842			15.11103	5.71318
2 Futures	1-m, 2-m	-0.00088	1.27315	-0.27227		6.26711	2.79411
	1-m, 6-m	-0.00171	1.23899	-0.23729		6.27232	2.97602
	1-m, 12-m	-0.00021	1.19336	-0.19315		6.28735	3.00006
	2-m, 6-m	-0.04079	5.06602	-4.02523		10.27413	9.37292
	2-m, 12-m	-0.01179	2.94860	-1.93681		9.65705	7.08414
	6-m, 12-m	0.01481	4.80979	-3.82460		9.57938	4.52846
3 Futures	1-m, 2-m, 6-m	-0.18008	0.60862	0.14966	0.42180		
	1-m, 2-m, 12-m	-0.19688	0.79086	0.24270	0.16333		
	1-m, 6-m, 12-m	-0.20496	0.64458	0.41532	0.14506		
	2-m, 6-m, 12-m	-0.21907	0.47364	0.59770	0.14773		